

Auditing Operating Systems & Security Controls

Operating system controls

- The operating system is important for;
 1. Translating high level instructions into machine readable code
 2. Distribute and manage resources among users and applications
 3. Task management through job scheduling and multi-programming

Control activities of OS

1. Protection from users Prevents user applications from gaining control or damaging the OS, ensuring continued operation and data integrity.
2. Protection of Users from Each Other: Ensures that one user cannot access, destroy, or corrupt another user's data or programs.
3. Protection of Users from Themselves: Prevents different modules within a user's application from interfering with each other, protecting the integrity of the data and programs.

Security weaknesses

1. employees are using weak passwords

Each weak password is a vulnerability a threat actor can exploit to get unauthorized access into the system.

2. some staff have access to sensitive data without proper authorization

All employees, including low ranking employees and potential spies gaining access to sensitive, mission critical company information.

3. no proper monitoring of system activities

Without proper monitoring a simple issue that was not identified long ago can cause ripple effects and create much more severe issues later

4. system was recently affected by a virus attack

Loss of company secrets, confidential customer data and financial, reputational and legal issues.

Audit Procedures

Access Privileges

- Review organizational policy for separation of incompatible functions and supporting reasonable security
- Examine whether users only have privileges required by their positions
- Check if employees go through thorough security clearance checks
- Ensure users' log on permissions match the tasks being performed.

Password Controls

- Ensure that all employees have password authentication.
- Check the strength of individual passwords.
- Check the encryption level of passwords at rest.
- Review password change procedures

Malicious Program Protection

- Examine whether procedures are available to educate employees on malware and analyze their effectiveness.
- Check whether all new software is verified before connected to the company intranet
- Check the policies on antiviral software for mandatory installation and regular update policies

Audit Trail

- Review who, what and when is recorded in the audit trail.
- Examine whether audit logs are tamper proof
- Review audit trail usage statistics.
- Check audit trail retention and archival policies.

Recommendations

1. Ensure all employees have strong unique passwords through an authentication server ex: RADIUS and promote the adoption of one time use passwords
2. Regularly audit access controls to detect improper authorization levels
3. Use automated monitoring programs to monitor audit logs constantly or as per company policy
4. Mandate up-to-date Antiviral software installations on employee devices, especially for BYOD instances.